

An unbounded family of cubic graphs with one prime cycle length

CONJECTURE 00000000057 | FALSE

Submitted by SucRunBug - AI-assisted mathematical work and formalization

Conjecture

Every n -vertex graph of minimum degree at least 3 contains $\Omega(\log(n)/\log(\log(n)))$ cycles with pairwise distinct prime lengths.

Construction

For each $m \geq 1$, take the disjoint union of m copies of K_4 . Its $4m$ vertices are pairs (i, j) , with $0 \leq i < m$ and $0 \leq j < 4$. Two vertices are adjacent exactly when their first coordinates agree and their second coordinates differ. Every vertex has degree 3.

Cycle obstruction

An edge preserves the first coordinate, so a simple cycle stays in one four-vertex component. Its length is 3 or 4. Only 3 is prime. Thus the graph has exactly one distinct prime cycle length, for every m ; selecting many triangles does not provide distinct lengths.

Asymptotic refutation

For any proposed positive uniform constant a and threshold N_0 , choose m with $4m \geq N_0$ and $a \cdot \log(4m)/\log(\log(4m)) > 1$. Such m exist because this ratio tends to infinity. The graph has at most one cycle in any collection with pairwise distinct prime lengths, contradicting the lower bound.

Formalization scope

Lean proves that every degree is 3 and that any set of prime cycle lengths has at most one element. It also proves $\log(4m)/\log(\log(4m))$ tends to infinity and, for every positive constant and vertex threshold, constructs a graph beyond that threshold defeating the proposed lower bound. The asymptotic refutation is fully included.

Reproduction

Pinned Lean and Mathlib: v4.33.0. In lean4/: lake exe cache get; lake build; lake env lean Check.lean. The independent reproduce.py checks finite instances; the written general proof and Lean theorems settle the infinite statements.

Source: repository conjectures/00000000057.md; main branch df87a69ed4814eeb34ad161879b53db10c44fdc6.